7.2 Multiplying Polynomials

Name _____ Period ____ Date _____

Algebra I 7.2 · Student sheet

I can…

- multiply a monomial and a polynomial.
- multiply two binomials using a table, the Distributive Property, or FOIL.
- multiply a binomial and a trinomial.

Vocabulary · fill in the missing word

Distributive Property — a(b + c) = ab + ac: _____ every term inside by the term outside

FOIL Method — First, Outer, Inner, Last: the four products when _____ two binomials

Area model (table) — a _____ split into boxes, one for each pair of terms

Part 1 · Area models

Fill in each area model. Write the area of each box, then add to get one polynomial.

(x + 3)(x + 2)	x	+3
x		
+2		

(x + 3)(x + 2) = _____

(2x + 1)(x + 4)	2x	+1
x		
+4		

(2x + 1)(x + 4) = _____

Which boxes were like terms? _____

Part 2 · Notes: fill in the blanks

To multiply a monomial and a polynomial, _____ the monomial to every term.

When I multiply powers with the same base, I _____ the exponents.

FOIL stands for _____, _____, _____, _____.

A binomial times a trinomial gives _____ products before I combine _____.

Part 3 · Examples (work them with the class)

Example 1

Find the product.

$-3x^2(2x^3 - 4x + 1)$

Example 2

Find the product.

$(x + 4)(x - 7)$

Example 3

Use a table to find the product.

$(3x - 2)(x + 5)$

Example 4

Use FOIL to find the product.

$(2y - 3)(4y - 1)$

Example 5

Find the product.

$(x + 2)(x^2 - 3x + 4)$

Example 6

A rectangle has length $(2x + 5)$ and width $(x - 1)$. Write a polynomial that represents its area.

Part 4 · Partner practice

Partner 1

Find the product.

$(x - 6)(x + 2)$

Partner 2

Find the product.

$(2n + 1)(n^2 - n + 3)$